Missing Eigenvectors

Consider $\boldsymbol{y}' = \boldsymbol{A}_\varepsilon \boldsymbol{y}$ with $\varepsilon \neq 0$.

$$\boldsymbol{A}_\varepsilon = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda + \varepsilon \end{pmatrix}$$

The distinct eigenvalues give two independent solutions.

$$\lambda_1 = \lambda \qquad \boldsymbol{V}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \boldsymbol{y}_{1,\varepsilon}(t) = e^{\lambda t}\boldsymbol{V}_1$$

$$\lambda_2 = \lambda + \varepsilon \quad \boldsymbol{V}_2 = \begin{pmatrix} 1 \\ \varepsilon \end{pmatrix} \quad \boldsymbol{y}_{2,\varepsilon}(t) = e^{(\lambda+\varepsilon)t}\boldsymbol{V}_2$$

As $\varepsilon \to 0$, $\boldsymbol{V}_2 \to \boldsymbol{V}_1$ and both solutions approach $e^{\lambda t}\boldsymbol{V}_1$.

A Second Solution in the Limit

For $\varepsilon \neq 0$, the scaled difference is also a solution.

$$\frac{\boldsymbol{y}_{2,\varepsilon}(t) - \boldsymbol{y}_{1,\varepsilon}(t)}{\varepsilon} = e^{\lambda t} \begin{pmatrix} (e^{\varepsilon t} - 1)/\varepsilon \\ e^{\varepsilon t} \end{pmatrix} \xrightarrow{\varepsilon \to 0} e^{\lambda t} \begin{pmatrix} t \\ 1 \end{pmatrix}$$

The first entry inside the vector tends to t by L'Hôpital's rule.

For $\boldsymbol{W} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, $(\boldsymbol{A}_0 - \lambda \boldsymbol{I})\boldsymbol{W} = \boldsymbol{V}_1$, so

$$\boldsymbol{y}_2(t) = e^{\lambda t}(t\boldsymbol{V}_1 + \boldsymbol{W})$$

solves $\boldsymbol{y}' = \boldsymbol{A}_0\boldsymbol{y}$. Its second entry is $e^{\lambda t}$, while $e^{\lambda t}\boldsymbol{V}_1$ has second entry zero.

Reduction of Order

The trick is to let the coefficients of $\boldsymbol{V}_1$ and $\boldsymbol{W}$ vary. For the limiting system $\boldsymbol{y}' = \boldsymbol{A}_0\boldsymbol{y}$, set

$$\boldsymbol{y}(t) = e^{\lambda t}\big(a(t)\boldsymbol{V}_1 + b(t)\boldsymbol{W}\big)$$

Substitute and cancel the terms from differentiating $e^{\lambda t}$. Since $(\boldsymbol{A}_0 - \lambda\boldsymbol{I})\boldsymbol{V}_1 = 0$ and $(\boldsymbol{A}_0 - \lambda\boldsymbol{I})\boldsymbol{W} = \boldsymbol{V}_1$, we get

$$a'(t)\boldsymbol{V}_1 + b'(t)\boldsymbol{W} = b(t)\boldsymbol{V}_1 \qquad \Longrightarrow \qquad a'(t) = b(t), \quad b'(t) = 0$$

Choose $b(t) = 1$ and $a(0) = 0$ to get $a(t) = t$. This gives the same second solution as the limit.

$$\boldsymbol{y}_2(t) = e^{\lambda t}(t\boldsymbol{V}_1 + \boldsymbol{W})$$